

Inclusive E-Mobility for a Just Transition

RULES OF ENGAGEMENT

Free Movement & Focus

Move between insight stations freely, spending time with those most relevant to your work - each station tells part of Ghana's E2&3W story

Interactive Participation

Where you see this icon, use the voting stickers and sticky notes to interact with the stations - add your insights, questions, and local context to sections, charts and data points



Connect & Contribute

Use evidence across stations to build comprehensive understanding and inform your contributions to group discussions

Access Detailed Information

Reach out to a CEDA representative for questions & clarifications, full reports, underlying data, methodology details, and collaboration opportunities

Stay Engaged

Contact CEDA team for next research phases, policy implications, partnership possibilities, and follow-up consultation opportunities

Your local knowledge and sectoral expertise are crucial for interpreting these findings and shaping Ghana's inclusive E2&3W transition strategy.

Exploring Ghana's Transition to E2&3Ws

Comprehensive Review and Gap Analysis (D1)

INSIGHT STATION 1

Project Scope

Focus Areas

E2&3W transport policy, market potential, infrastructure, & socio-economic impacts

Geographic Coverage

National analysis; regional segmentation focusing on Greater Accra Ashanti & Northern

Stakeholder Scope

Government ministries, private sector, transport operators, development partners & CSOs

Analytical Framework

Comparative case studies with India, Kenya, Rwanda & China) with GESI lens & PEA

Purpose of D1

Gap Identification

Document critical policy, regulatory & market gaps hindering E2&3W adoption in Ghana

Baseline Establishment

State analysis of e-mobility landscape, vehicle registration trends & institutional frameworks

International Benchmarking

Compare Ghana's approach against proven international models & identify best practices

Stakeholder Validation

Engage diverse stakeholders to pressure-test findings & gather implementation insights

Key Themes from D1

Market Dynamics & Opportunity

Policy & Regulatory Gaps

Gap Identification

Social Inclusion Challenges

Validate & Mobilise

Stakeholder Input

Review findings for accuracy, relevance, and implementation feasibility. Identify missing perspectives & priority intervention areas.

Next Phase Focus

D1 insights inform subsequent research phases covering economic viability, infrastructure constraints, & manufacturing potential assessments.

Need more clarity or further information?
Speak to a CEDA Representative!

Legal & Institutional Framework

INSIGHT STATION 2

LEGAL & POLICY TIMELINE

2004 Road Traffic Act (Act 683)

Foundation legislation for vehicle registration, licensing, and transport regulation

Impact: Core regulatory framework established

2012 Road Traffic Regulations (L.I. 2180)

Detailed regulations for transport operations, including "okada" commercial ban

Impact: Major barrier for E2W commercial operations

2019 Automotive Development Policy

Focus on 4-wheeled vehicle assembly, limited E2&3W provisions

Gap: Missed E2&3W manufacturing opportunity

2020 Road Traffic Amendment Act (Act 1014) & Revised Transport Policy

Updated vehicle standards and roadworthiness; lacks EV-specific provisions

Gap: No electric vehicle differentiation despite environmental benefits

2022 GSA Act (Act 1078) & Energy Transition Framework

Enhanced standards authority; Energy Transition Framework identifies E2&3Ws as "quick wins"

Opportunity: Clear E2&3W policy recognition

2023 National Electric Vehicle Policy & Energy Transition Investment Plan

First dedicated EV policy framework; Emphasis on E2&3Ws enhancing mobility for low-income populations

Progress: Policy intent established, implementation gaps remain

2025 EPA Amendment Act

Expanded EPA authority for EV battery recycling and charging infrastructure oversight

Progress: Environmental regulation framework enhanced

POLITICAL ECONOMY ANALYSIS

New Government Transition

New government (inaugurated Jan 7, 2025) committed to fiscal consolidation & IMF program adherence, with significant appointments across transport & energy sectors affecting institutional capacity

Ministry of State for Climate Change & Sustainability

New Ministry of State under Presidency created for Climate Change & Sustainability, gaining prominence and offering potential entry point for strategic climate change program engagements

"Okada" Legalisation Plans

President John Mahama announced plans to legalize & regulate "okada" (motorcycle taxis) through amendments to the Roads Act, introducing licensing, training, safety rules, ID & age requirements, & mandatory helmets & plates

Regional Political Economy Variations

Northern regions show high 2&3W acceptance with strong cultural adoption, contrasting with southern regions facing regulatory resistance & cultural barriers

Sub-National Influence

District assemblies, traditional authorities, & regional coordinators emerge as critical decision-makers for community acceptance, wielding substantial influence over e-mobility initiative success

Institutional Coordination

Transport and energy policies operating in silos limit strategic alignment, contrasting with more centralized coordination in peer countries like Rwanda & India

Support for Climate Action Goals

Commitment to update government vehicle fleet with EVs for ministers & heads of public institutions to support its climate action goals, reduce emissions & promote sustainable transport

POLICY & REGULATORY GAPS

Commercial Operations Ban

"Okada" restrictions prevent legal E2W commercial deployment despite environmental benefits

E2&3W-Specific Standards

No tailored safety, battery, & performance standards for light electric vehicles and E2&3Ws

Fragmented Implementation

Policy coordination gaps between transport, energy, & environmental agencies

Informal Sector Integration

No regulatory framework for widespread informal 2&3W operators a large number of youth riders and informal mechanics

Informal Sector Integration

Limited EV-specific import duties & tax frameworks compared to regional peers

Need more clarity or further information?
Speak to a CEDA Representative!



Legal & Institutional Framework

INSIGHT STATION 2: INTERACTION BOARD

Question 1

How has the Okada ban specifically affected your community/business operations?

Write on sticky note and place below

Place Your Sticky Note Responses Here

Question 2

What would successful Okada legalization look like in practice for your sector?

Write on sticky note and place below

Place Your Sticky Note Responses Here

Which political economy event most influenced socio-economic standing in your community?

Place your colored sticker next to each stage

Blue - Low: Event had marginal or background impact

Orange – Medium: Event had a clear effect on direction or pace of progress

Green – High: Event was pivotal, directly determining success or failure

New Government Transition

Place Your Colored Stickers Here.

Ministry of State for Climate Change & Sustainability

Place Your Colored Stickers Here.

“Okada” Legalisation Plans

Place Your Colored Stickers Here.

Sub-National Influence

Place Your Colored Stickers Here.

Regional Political Economy Variations

Place Your Colored Stickers Here.

Institutional Coordination

Place Your Colored Stickers Here.

Support for Climate Action Goals

Place Your Colored Stickers Here.

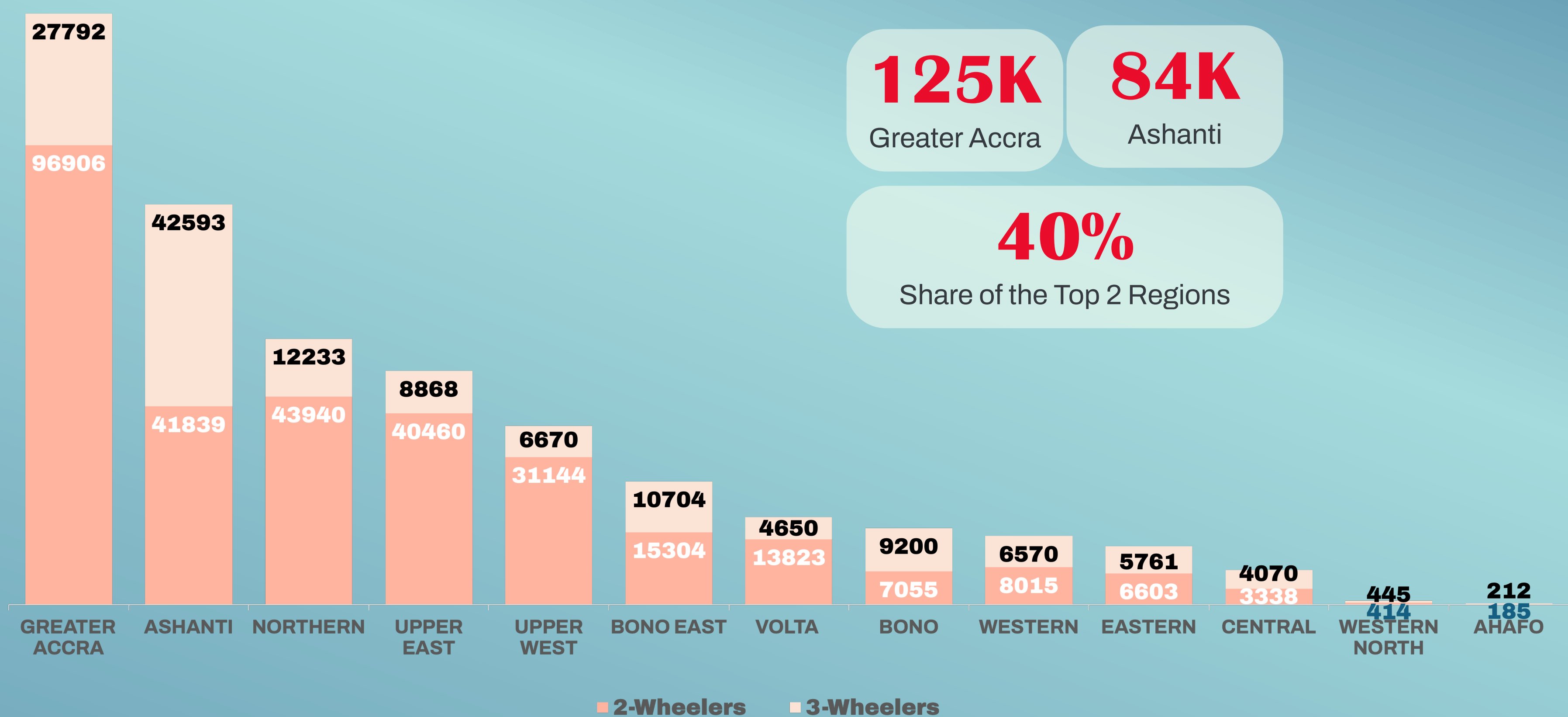
Need more clarity or further information?

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Regional 2&3W Utilisation Patterns

INSIGHT STATION 3

Urban Concentration vs Rural Demand



Regional Concentration Highlights Infrastructure Priorities

Urban Concentration: Greater Accra and Ashanti account for 40% of all registrations, indicating urban economic hubs where charging infrastructure investment would yield maximum impact.

Northern Regions Potential: Northern, Upper East, and Upper West show substantial motorcycle usage, suggesting strong demand for affordable electric alternatives in these economically under-served areas.

Need more clarity or further information?
Speak to a CEDA Representative!



Regional 2&3W Utilisation Patterns

INSIGHT STATION 3 : INTERACTION BOARD

Question 1

Have you ever used an Okada to get somewhere a car couldn't reach (traffic, no car road, cost)? Was this bike electric?

Place Your Colored Stickers Below:

Yes, An Electric Bike

Yes A Regular Bike

No, Haven't Needed a Bike

Question 2

What would convince people in your region to switch from petrol to electric 2&3Ws?

Write on sticky note and place below

Place Your Sticky Note Responses Here

**Should urban areas
(Greater Accra &
Ashanti) get priority
investment over rural
regions?**

Yes

*Place Your Colored Stickers
Here. Any Color!*

No

*Place Your Colored Stickers
Here. Any Color!*

No

*Place Your Colored Stickers
Here. Any Color!*

**Do you think the
Northern regions have
enough 2&3W demand to
justify major E2&3W
programs?**

Yes

*Place Your Colored Stickers
Here. Any Color!*

No

*Place Your Colored Stickers
Here. Any Color!*

Not Sure

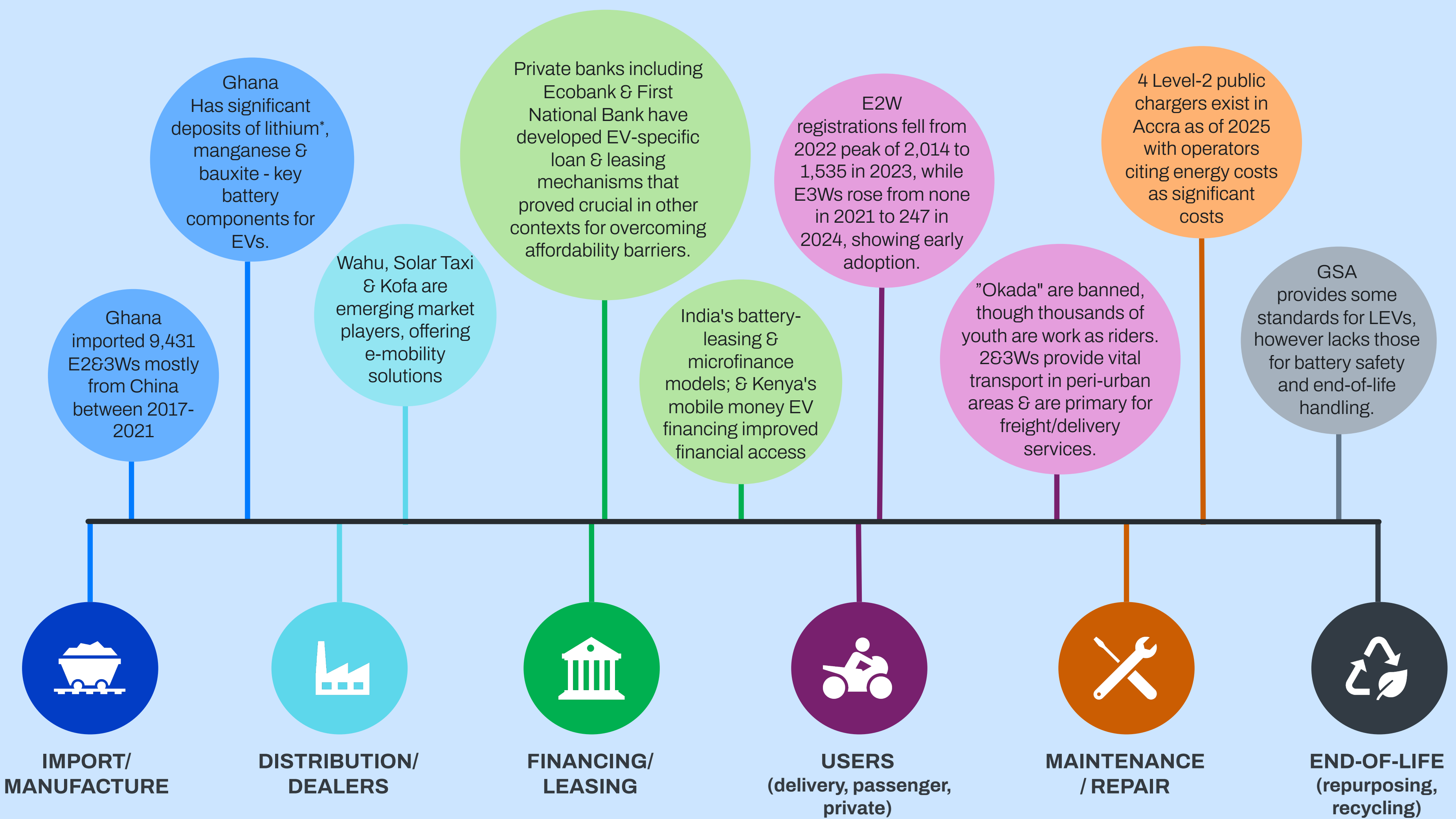
*Place Your Colored Stickers
Here. Any Color!*

Need more clarity or further information?

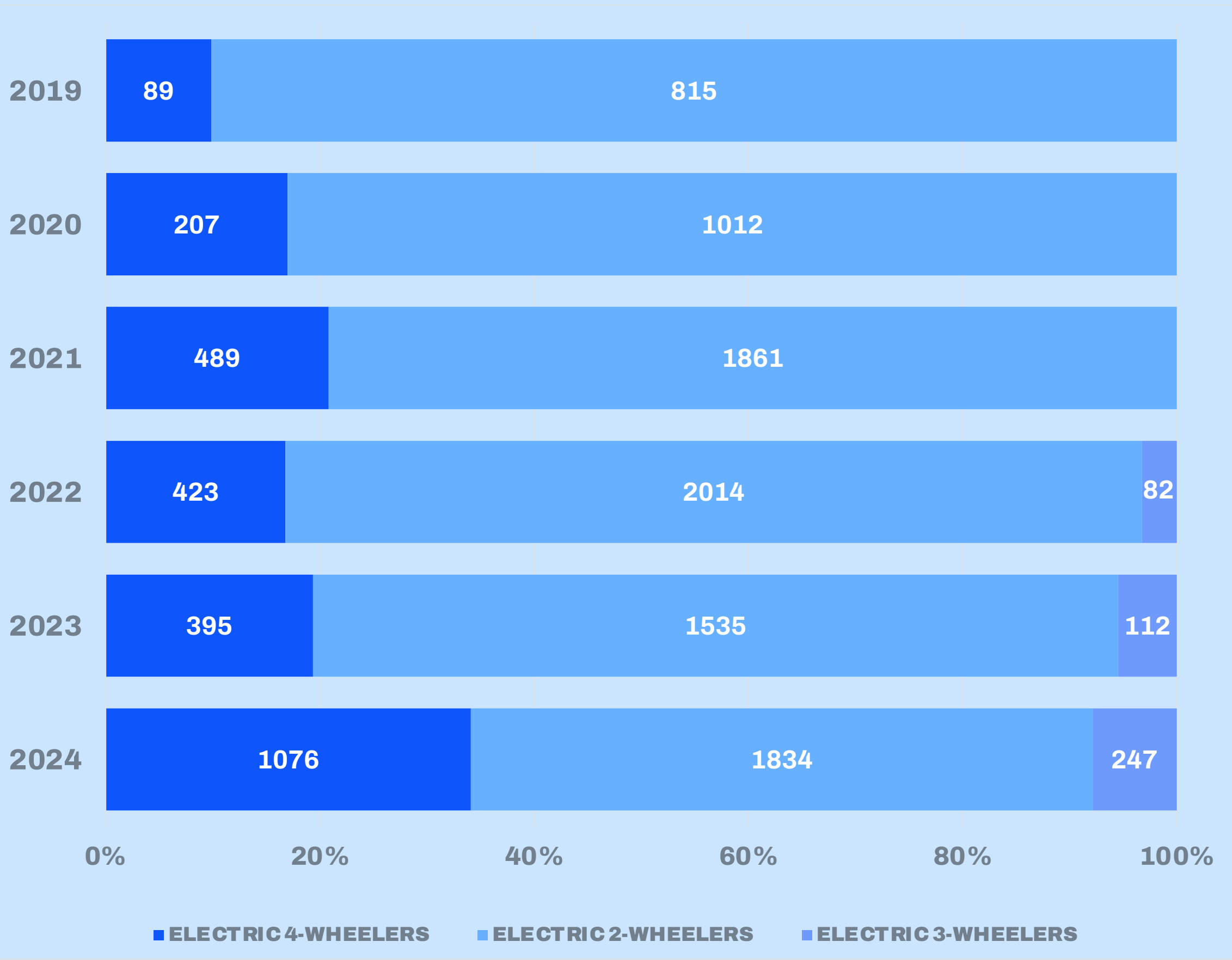
Speak to a CEDA Representative!

E2&3W Value Chain

INSIGHT STATION 4



*Ewoyaa Lithium Project, Ghana's first lithium-producing mine, is set to commence production in 2025.



Early-Stage Electric Adoption with Accelerating Growth

Electric 2&3-wheeler registrations increased **155% from 2019-2024**, reaching **2,081 units despite** representing only **1.96% market penetration**.

Electric 4-wheelers show dramatic growth with **1,109% increase**, indicating significantly growing consumer acceptance.

The emergence of electric 3-wheelers from 2022 onwards **signals market diversification, growing commercial operator interest in alternative fuel technology and readiness for broader EV adoption** across Ghana's transport sector.

Need more clarity or further information?
Speak to a CEDA Representative!



E2&3W Value Chain

INSIGHT STATION 4 : INTERACTION BOARD

Question 1

If you needed to buy an electric 2 or 3-wheeler today, what would be your biggest concern?

Cost? Charging? Maintenance? Quality? - write on sticky note and place below

Place Your Sticky Note Responses Here

Question 2

Which part of the value chain offers the best business opportunity for local entrepreneurs in your area?

Write on sticky note and place below

Place Your Sticky Note Responses Here

Which value chain stage needs the most urgent development?

Place your colored sticker next to each stage

Blue – Low: Can be addressed gradually; limited immediate risk.

Orange – Medium: Needs attention soon; delays could worsen challenges.

Green – Urgent: Immediate intervention required; critical risks if not acted on quickly.

**Import/
Manufacture**

Place Your Colored Stickers Here

Distribution

Place Your Colored Stickers Here

Financing

Place Your Colored Stickers Here

Users

Place Your Colored Stickers Here

**Maintenance/
End-of-Life**

Place Your Colored Stickers Here

Need more clarity or further information?

Speak to a CEDA Representative!

Comparative Gap Analysis

INSIGHT STATION 5

Gap Analysis Framework

This analysis identifies critical gaps between Ghana's current E2&3W approach and successful international transitions. Case study countries (Rwanda, Kenya, India, Norway, China, South Africa) provide benchmarks across policy incentives, regulatory frameworks, infrastructure deployment, and financing ecosystems.

Gap Category	Ghana's Current Status	Best Practice Examples	Gap Severity*
Policy Framework	- No E2&3W-specific incentives - No concrete EV adoption targets - Policies operate in silos - Transport vs EV policy conflicts	Norway: 0% VAT, free tolls, parking incentives → 93% EV adoption China: Strategic industrial policy with R&D support Kenya: Fiscal-first approach suitable for import-dependent markets	
Regulatory Framework	"Okada" commercial ban prevents E2W deployment - No E2&3W safety standards - No battery recycling regulations - Limited EV classification system	China: Early e-bike classification, city-level ICE bans India: Legal recognition of e-rickshaws in 2015	
Infrastructure	- Only 4 Level-2 public chargers in Accra - Limited battery swapping networks - Grid reliability issues in regions - Inadequate local assembly capacity	China: Comprehensive technical standards, integrated grid planning Rwanda: Solar backup integration, coordinated public-private model Kenya: Targeted private-sector charging approach	
Financing Ecosystem	- Few EV-specific loan products - Limited microfinance options - High upfront costs vs ICE alternatives - No battery leasing mechanisms	India: Battery leasing & microfinance models Kenya: Mobile money EV financing Rwanda: Battery-as-a-service model	



Place your colored sticker next to each gap category: **Blue** for Critical gaps requiring immediate action, **Orange** for Moderate gaps needing medium-term attention, and **Green** for Minor gaps that can be addressed long-term.



KENYA

Eliminated import duty/VAT on EVs & mainstreamed mobile money financing. **Key lesson:** Fiscal-first approach works for import-dependent markets with innovative financing.



RWANDA

Zero-tax regime + solar backup integration. **Key lesson:** Centralized coordination between government, development partners & private sector accelerates transition.



INDIA

Battery leasing, microfinance models & E3W legalization. **Key lesson:** Fiscal-first approach works for import-dependent markets with innovative financing.

Need more clarity or further information?

Speak to a CEDA Representative!



Comparative Gap Analysis

INSIGHT STATION 5 : INTERACTION BOARD

Question 1

Which international example (Kenya, Rwanda, India) would work best in your specific region of Ghana?

Write region and country choice with reason on sticky note

Place Your Sticky Note Responses Here

Place Colored Sticker Next to each Recommendation to Vote on Priority:

Blue – Low Priority: foundation work but not immediately critical for E2&3W adoption. **Orange – Medium Priority:** delays could worsen challenges and slow down the transition significantly. **Green – Urgent:** critical risks if not acted on quickly

Establish legal framework for electric motorcycle taxis and phase out "okada" ban for EVs

Following India's e-rickshaw legalization model, create regulatory pathways for commercial E2&3W operations with safety standards, licensing requirements, and training programs.

Place Your Colored Stickers Here

Implement comprehensive tax incentives similar to Kenya's approach

Eliminate import duties and VAT on E2&3Ws, introduce purchase subsidies for low-income users, and create preferential tax treatment for E2&3W businesses and charging infrastructure.

Place Your Colored Stickers Here

Deploy solar-powered charging stations and pilot battery swapping networks

Following India's e-rickshaw legalization model, create regulatory pathways for commercial E2&3W operations with safety standards, licensing requirements, and training programs.

Place Your Colored Stickers Here

Create mobile money-integrated financing and battery-leasing models

Develop EV-specific microfinance products, battery-as-a-service schemes, and fleet financing for commercial operators, drawing from India and Kenya's successful models.

Place Your Colored Stickers Here

Multi-Stakeholder Coordination Platform

Create coordinated platform bringing together government agencies, private sector, academia, and civil society following Kenya's EMAK model but with government leadership like Rwanda's approach.

Place Your Colored Stickers Here

Got More Recommendations?

Put a Sticky Note Anywhere on this Station!

Gender Equality & Social Inclusion

INSIGHT STATION 6

Our GESI Framework for E2&3 Mobility

Ghana's National Electric Vehicle Policy (2023) emphasises a "just, equitable, & inclusive transition, ensuring that the adoption of EVs benefits all segments of society." This analysis identifies specific groups at risk of exclusion & the barriers they face across social, economic, & institutional dimensions using the World Bank's Social Inclusion Assessment Tool (SIAT) framework.



Women

Social Norms:

Female riders labeled as "reckless" or "masculine" • North-South usage divide reflects deeper societal structures • Cultural norms limit mobility & economic independence

Economic Barriers:

Limited access to credit • Lower disposable income • Restricted economic opportunities

Policy Gaps:

No gender-responsive incentive structures • Limited representation in policy design



Youth & Informal Workers

Employment Issues:

Tens of thousands work as motorcycle taxi riders • Jobs at risk from transition without reskilling

Skills Gaps:

Limited technical training for EV maintenance • No pathways for upgrading informal work

Financial Access:

Cannot afford vehicle ownership • Need rental/leasing models



Persons with Disabilities

Design Barriers:

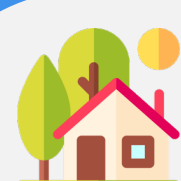
Inaccessible vehicle designs • No adaptive features for different disabilities

Infrastructure Issues:

Charging stations not accessible • Poor road/pathway conditions

Policy Oversight:

Disability inclusion not prioritised in EV policy • Limited accessibility standards



Rural Communities

Infrastructure Deficit:

Overlooked in transport infrastructure development • Limited charging infrastructure

Service Access:

Reduced access to healthcare, education & markets • Geographic isolation

Economic Impact:

Perpetuates cycles of rural poverty • Limited livelihood opportunities



Low-Income Populations

Affordability Barriers:

High upfront EV costs • Risk of being left further behind in transition

Financial Exclusion:

Limited access to formal credit • No EV-specific financing products

Poverty Cycles:

Unable to access improved mobility benefits • Trapped in existing transport costs

Need more clarity or further information?

Speak to a CEDA Representative!



Gender Equality & Social Inclusion

INSIGHT STATION 6 : INTERACTION BOARD

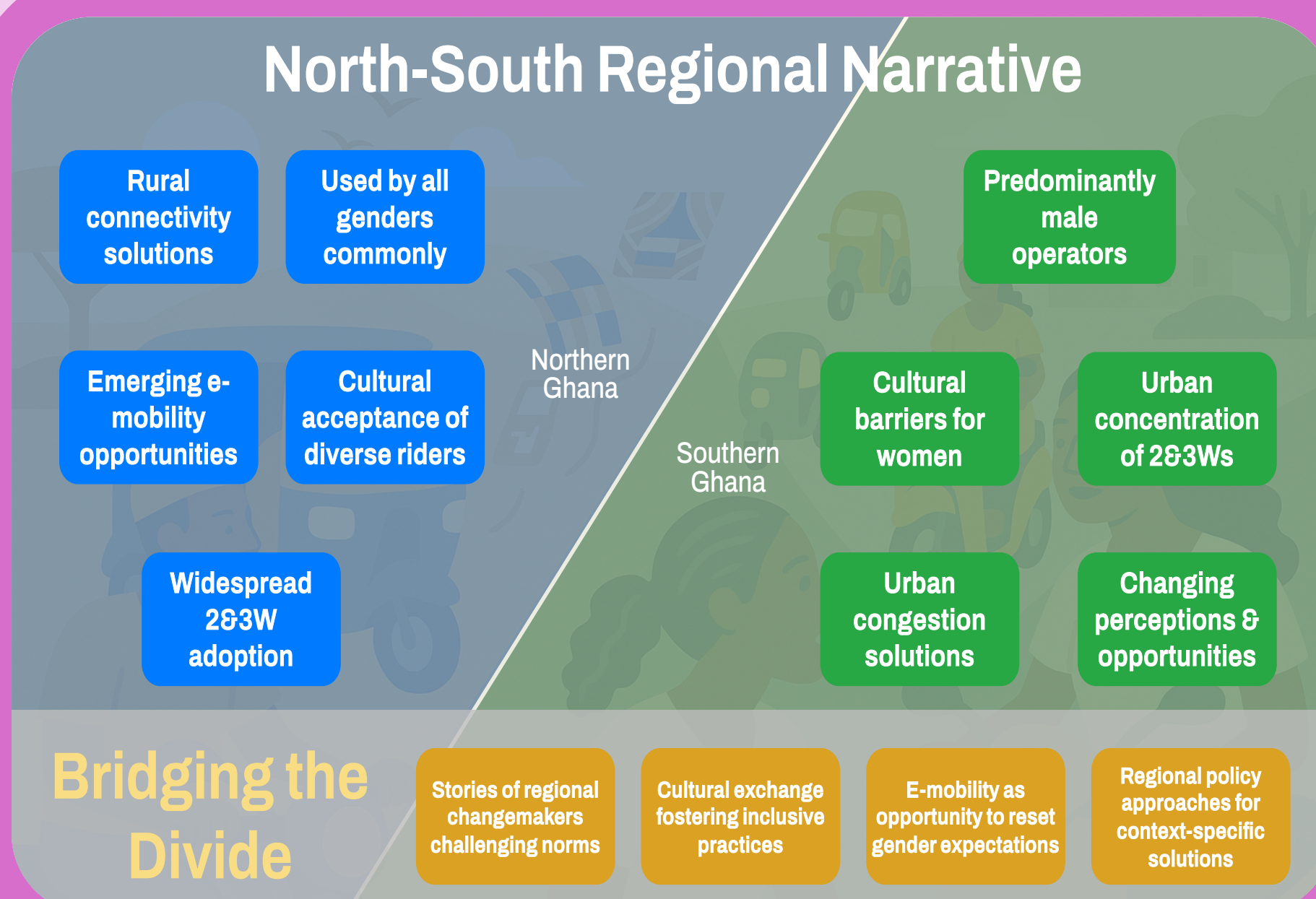
Question 1

**In your community, what prevents women from using motorbikes?
Or what enables women to use them?**

Write on sticky note and place below

Place Your Sticky Note Responses Here

North-South Regional Narrative



Northern Ghana: Women commonly ride motorcycles with motorcycles representing up to 98% of registered vehicles in some northern cities.

Southern Ghana: Traditional gender roles cast riding as a "male activity," creating social stigma that limits women's participation in motorcycle transport and related economic opportunities.

Question 2

How well does the North-South divide description match your experience?

Blue: Very Accurate **Orange: Somewhat True** **Green: Untrue**

Place Your Colored Stickers Here

You can also place sticky notes here to share insights

Need more clarity or further information?

Speak to a CEDA Representative!